Introduction to Computational Methods for Digital Humanists

VU MA Digitale Geisteswissenschaften, University of Graz, Summer term 2027

Berkeley, October 1, 2026

Dear participants,

I look forward to welcoming you to my course in the summer semester of 2027 and hope that we will have an inspiring and productive semester together.

Computational methods are a much-discussed subject in the digital humanities. In this introductory course, we will work through the core computational methods of the field along the steps of a research project: building a corpus from text, image or audio data, preparing and enriching it with metadata, annotating it manually and automatically, evaluating and analyzing the results, and presenting your findings. Throughout, we will reflect on the choices behind every step.

The course is a VU (Vorlesung mit Übung), so it is hands-on: every session pairs short input with practical work, and the emphasis is on examples of application and the completion of a group project. You will try out various digital tools and algorithms written in Python or accessible through a user-friendly interface. It is important that you are able to use the required infrastructure and understand what is happening as you apply it – so, please feel free to contact me and use the Moodle forum to communicate with your peers if you encounter any challenges.

Since we have only a limited number of synchronous sessions, there is neither time nor capacity to teach the basics of programming. I therefore encourage you to familiarize yourself with the files and tutorials on our Moodle page, to check that your computer can run a current version of Python, and to get a first footing before we start.

Please use the Moodle forum for any questions about the course content, its organization, programming problems, installation issues, and anything else. We benefit most when we use everyone's knowledge and help each other, so that we build an engaging, collaborative learning environment together.

This is a continuously assessed course (prüfungsimmanent), which means your grade comes from your work across the whole semester rather than from a single final exam. Attendance is mandatory, preparation matters, and active participation is part of your assessment. You may miss up to two sessions for important reasons if you let me know in advance.

The course is worth 4 ECTS, about 100 hours of work.

Your grade is based on three parts:

- Active participation and preparation of the sessions (25%)
    - do the preparation exercises and the reading
    - join the discussions, help each other, and take part in the shared learning process
- A group project (50%)
    - formulate a research question
    - build a small, rights-aware corpus and enrich it with metadata
    - operationalize the phenomenon you want to study and write annotation guidelines

    - create manual and automatic annotations and evaluate them
    - analyze and interpret your results
    - present your project as a poster (deadline: June 20, 2027) in a poster slam and gallery walk with Q&A
- An individual portfolio in three parts (25%)
    - Part 1: research question and corpus description (deadline: April 13, 2027)
    - Part 2: operationalization and annotation guidelines (deadline: June 8, 2027)
    - Part 3: reflection on your participation and learning process (deadline: July 15, 2027)
    - max. 10 pages in total, according to the instructions in *How to prepare the portfolio*
    - the guidelines may be written as a group; since you are individually responsible for your portfolio, adapt the shared text as you see fit

Course learning objectives

After successful completion of this course, you will know…

- content-oriented:
    - how to identify and formulate research questions in the digital humanities
    - how to build, document and prepare a small corpus of text, image or audio data in a rights-aware way
    - how to get from a humanities phenomenon to its operationalization and write annotation guidelines
    - how to annotate data manually and automatically, with pre-trained models and large language model systems
    - how to evaluate annotations and reflect on bias and the theory-dependence of computational categories
- skill-oriented:
    - how to plan, organize, and carry out a group project
    - how to work with Python, Jupyter notebooks, spaCy, transformer models, and local open-weights large language model systems
    - how to design a scientific poster and present research (poster slam and Q&A)
    - how to use questions to reflect on one's own work and learning progress

I am excited to accompany you on your learning journey and look forward to seeing you at the first session!

Best wishes,
Svenja Guhr